Introducing the hypothesis of demographic optimality in forests

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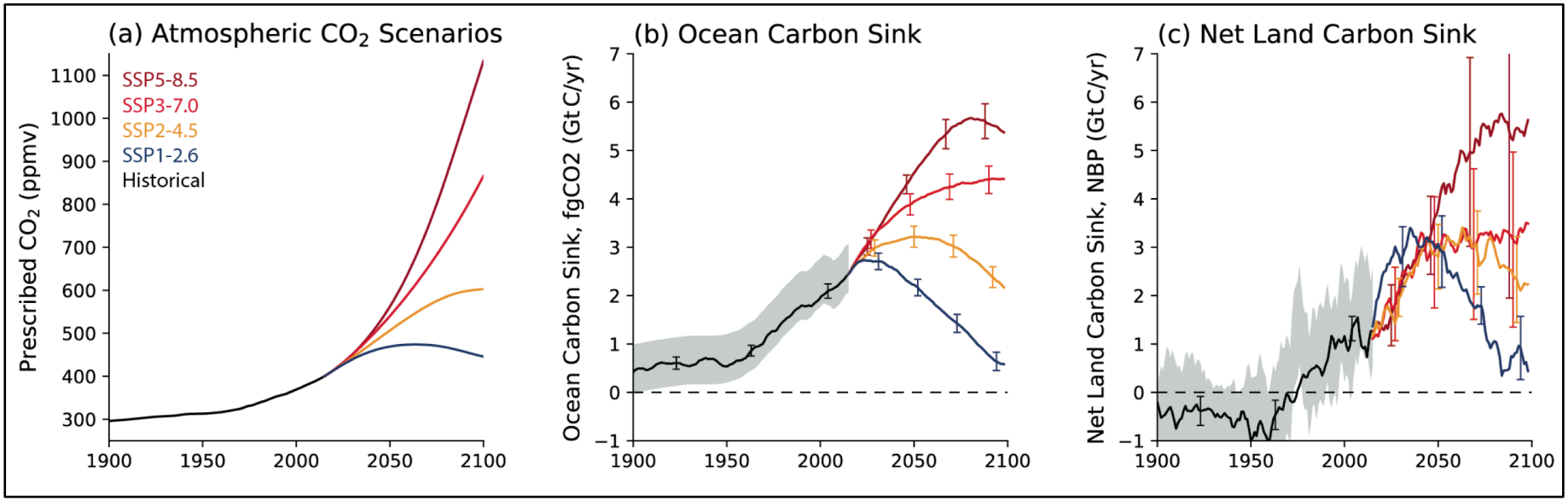
Projections of changes in global carbon sinks to 2100



IPCC AR6 WG1, Figure 5.25

Uncertainties in the future land carbon sink are still huge and are partly due to the representation of vegetation in Earth System/climate models……

Some Background

- Vegetation models in ESMs have to date been broad-brush and top-down (e.g. ignoring forest demography that is potentially important to predict changes in vegetation carbon in response to climate and land-use changes).
- New models that attempt to model forest demography are almost invariably more complex (and therefore have additional uncertain parameters).
- To derive analytical solutions (for comparison to numerical models), we initially made some heroic simplifications in RED.
- To our surprise, the braver the simplifications, the more our solutions looked like real forests…..☺

Governing Demographic Continuity Equation

$$\frac{\partial n(M,t)}{\partial t} + \frac{\partial}{\partial M}\{n(M,t)g(M,t)\} = -\gamma(M,t)\,n(M,t)$$

Growth-rate of Tree of mass *M*

Mortality-rate of Tree of mass *M*

- A continuity equation for tree number density ***n(M,t)***, with a sink term due to tree mortality and source term due to seedling production.
- Implicitly assumes that age-dependencies of growth or mortality can be approximated by mass-dependencies.

Simplifications

- We assume that net tree growth-rate (=NPP-local litterfall) depends on tree-mass to a power:

$$g = g_0 \left\{ \frac{m}{m_0} \right\}^b$$

- We apply Occam's razor and assume initially that mortality rate is independent of mass.

- We can estimate the exponent ***b*** from forestry data (for consistency with Metabolic scaling theory ***b***~3/4)

Steady-state solutions for non-zero mortality

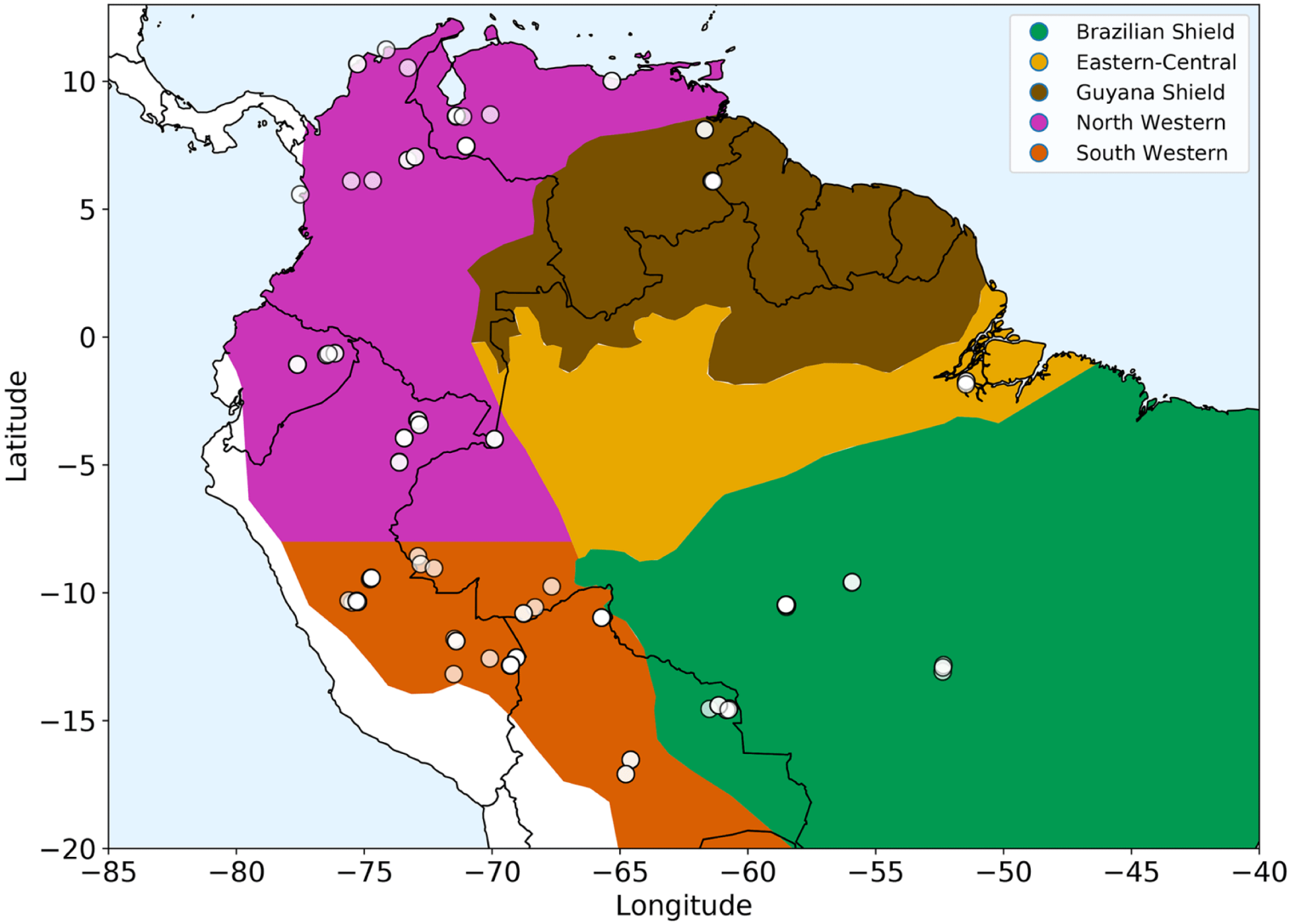
- The steady state solution is a Weibull distribution:

$$\frac{n}{n_0} = \left(\frac{m}{m_0}\right)^{-b} \exp\left\{\frac{\gamma m_0}{g_0(1-b)}\left[1-\left(\frac{m}{m_0}\right)^{1-b}\right]\right\}$$

- Reduces to a ‘self-thinning’ power law profile if mortality is negligible (note: this explanation of self-thinning is the simplest that we know of).

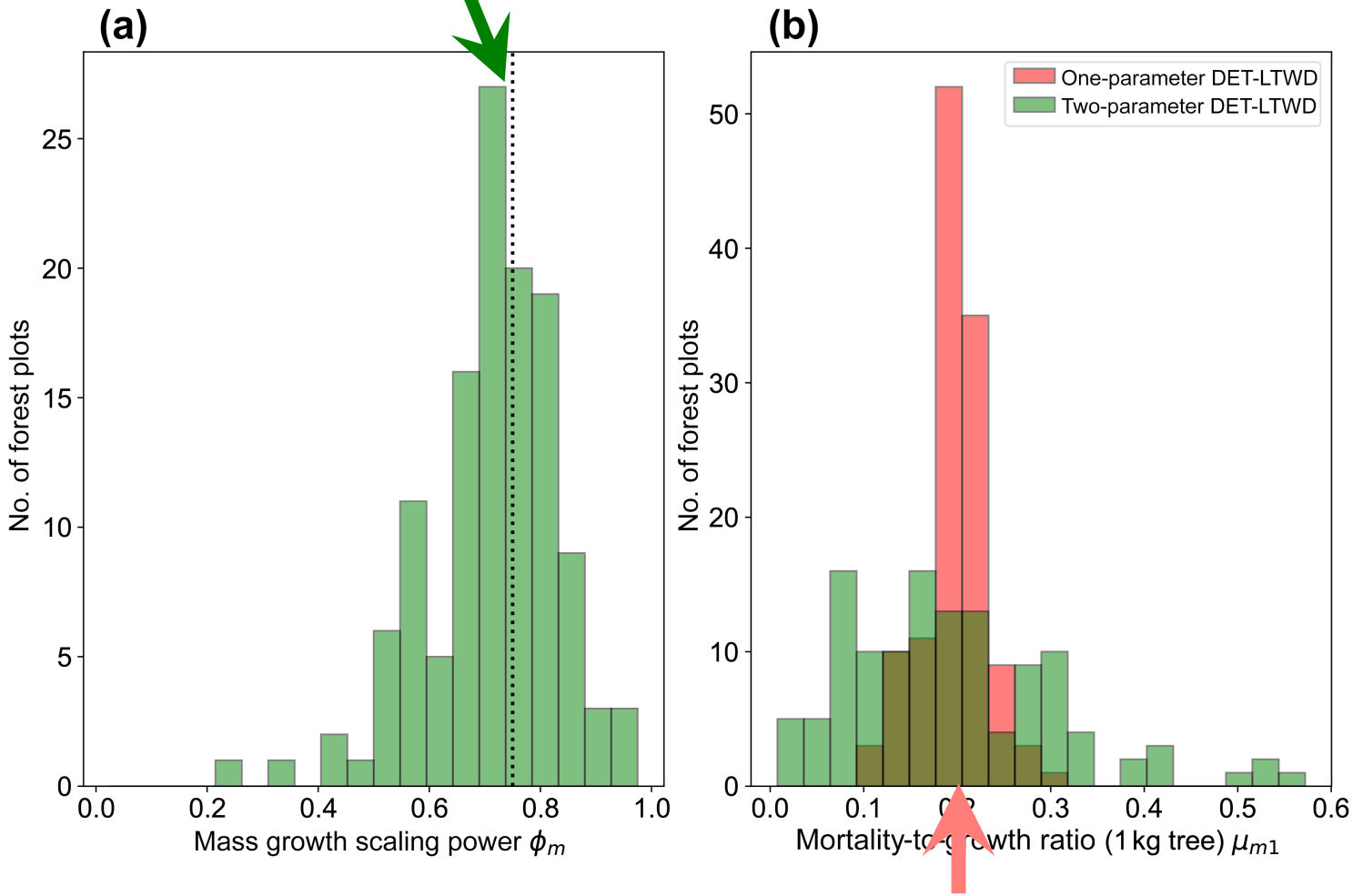
(Muller-Landau et al., 2006; Moore et al., 2018, 2020).

Tests against Forest Inventory data in Amazonia (RAINFOR)



Moore et al. (2020)

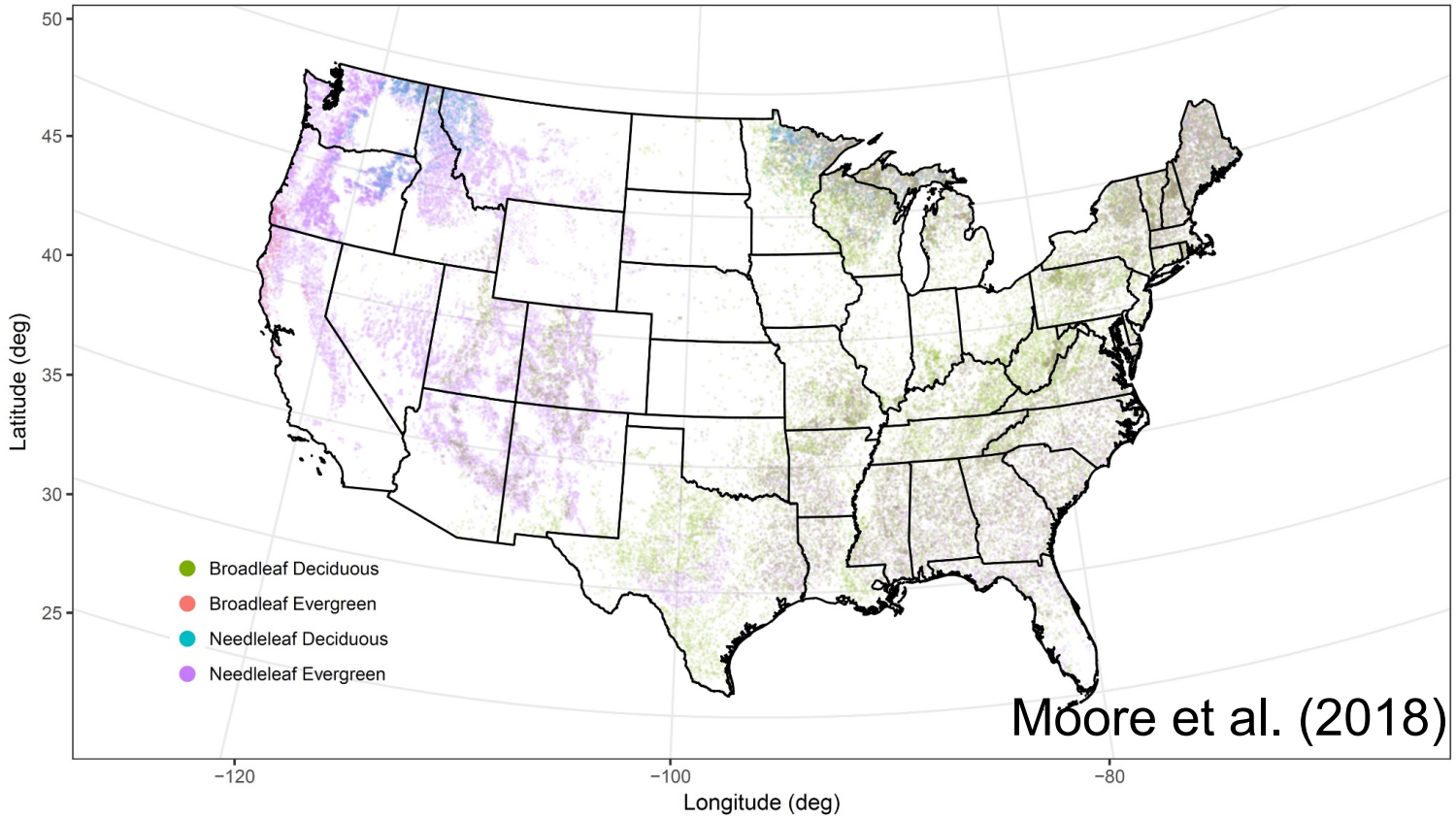
Best-fits to RAINFOR sites cluster around b=3/4 (consistent with MST!)



For b=3/4, the ratio of mortality to growth across sites is clustered in the range μ_1 ~ 0.2 to 0.25.

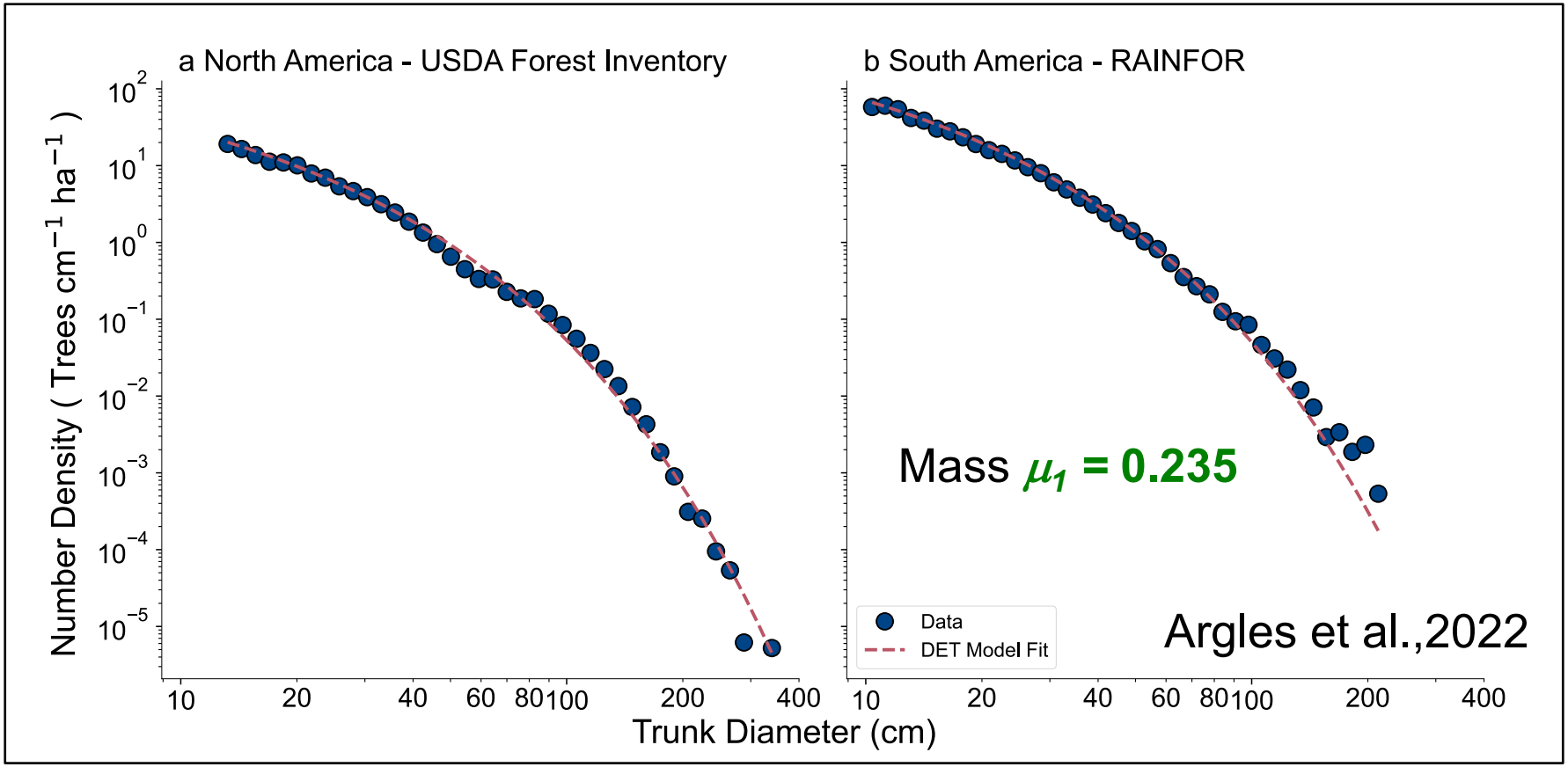
Moore et al. (2020)

We find similarly good fits to forest inventory data for N. America
(USDA Forest Inventory Data)



Trunk diameters for 1.6 million trees across 61 species (Zhu et al., 2012)

Evaluation of Demographic Equilibrium Theory with Metabolic Scaling Theory (b=0.75) for tree growth-rates



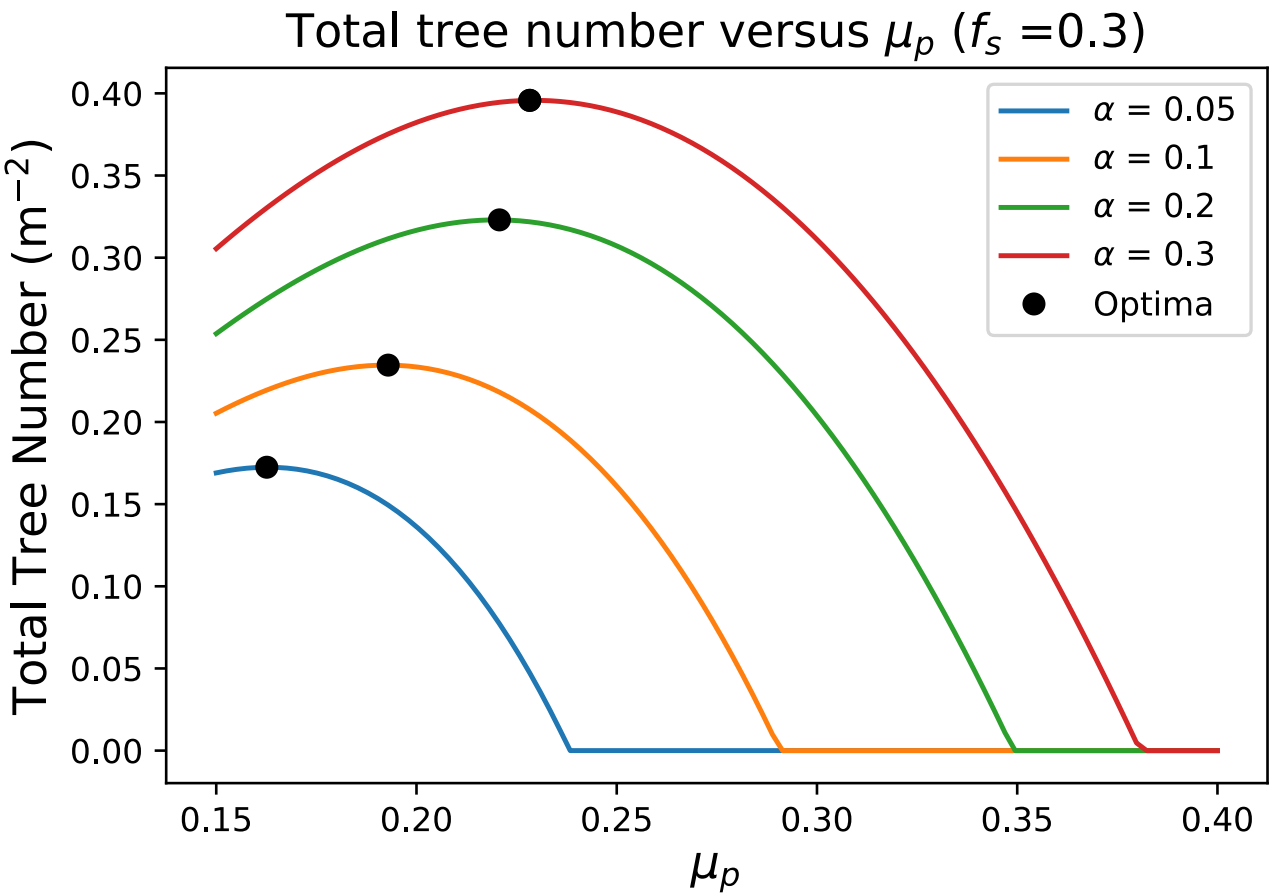
Similar best fit values for the ratio of mortality to growth!

Why might the ratio of mortality to growth (μ_1) be similar across very different forests?

- *Darwinian argument*: A species will dominate if it has the largest possible number of individuals that survive to reproduce.

- So does the observed value of μ_1 maximize the total tree number density?

Total tree number has a maximum as a function of the ratio of mortality to productivity (μ_p), for a given allocation of total assimilate to reproduction (α)



← *Mortality low so larger trees dominate, less trees overall*

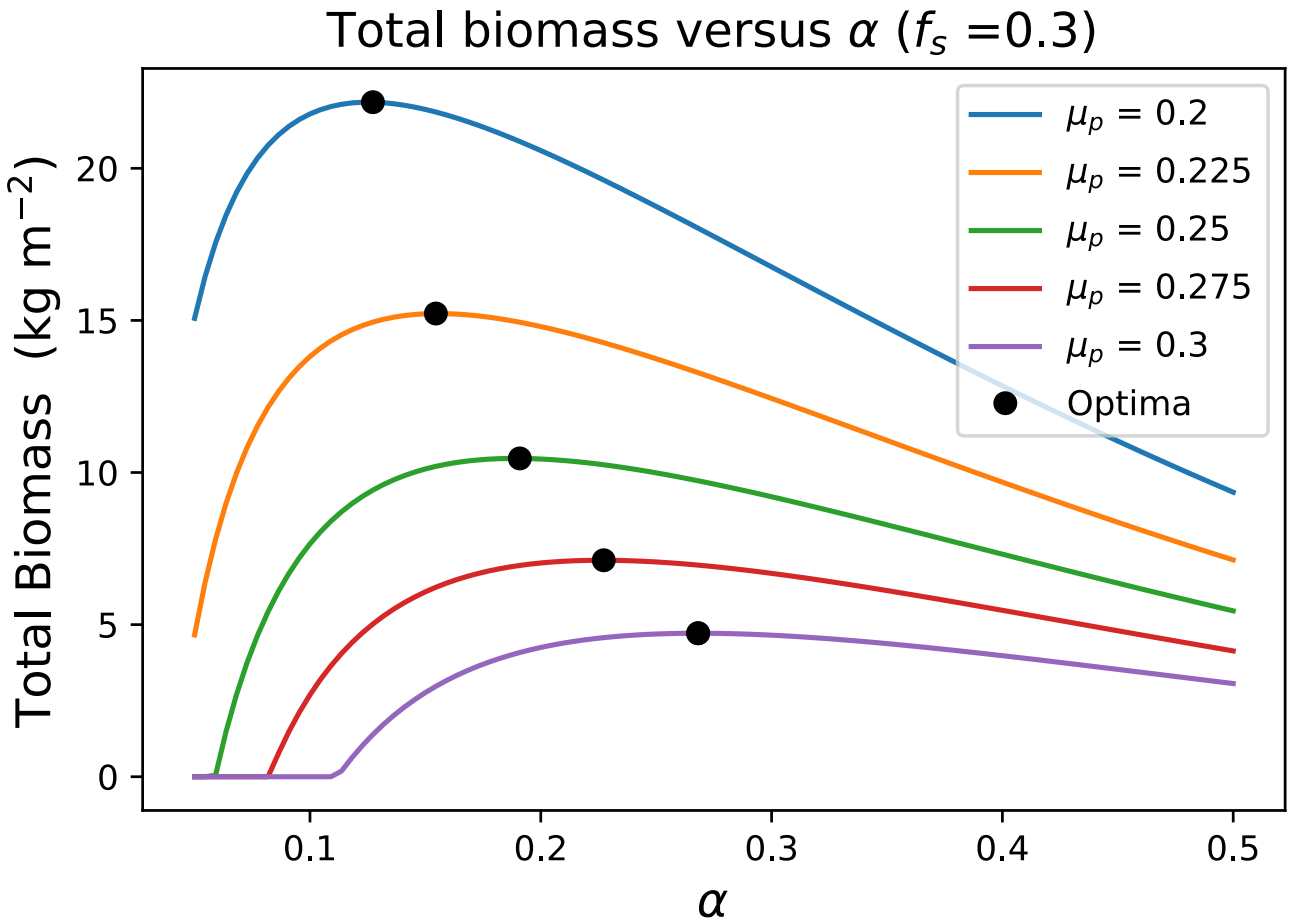
Mortality higher so fewer trees survive →

Why wouldn't a species devote a higher fraction of assimilate to reproduction (α) ?

- *Light-competition argument*: A species needs to also maximise biomass(and therefore height) as this is required to compete for light with other species.

- So how does total biomass vary with α ?

Total biomass tree has a maximum as a function of the allocation of total assimilate to reproduction (α), for a given ratio of mortality to productivity (μ_p)



Larger α values imply smaller overall biomass →

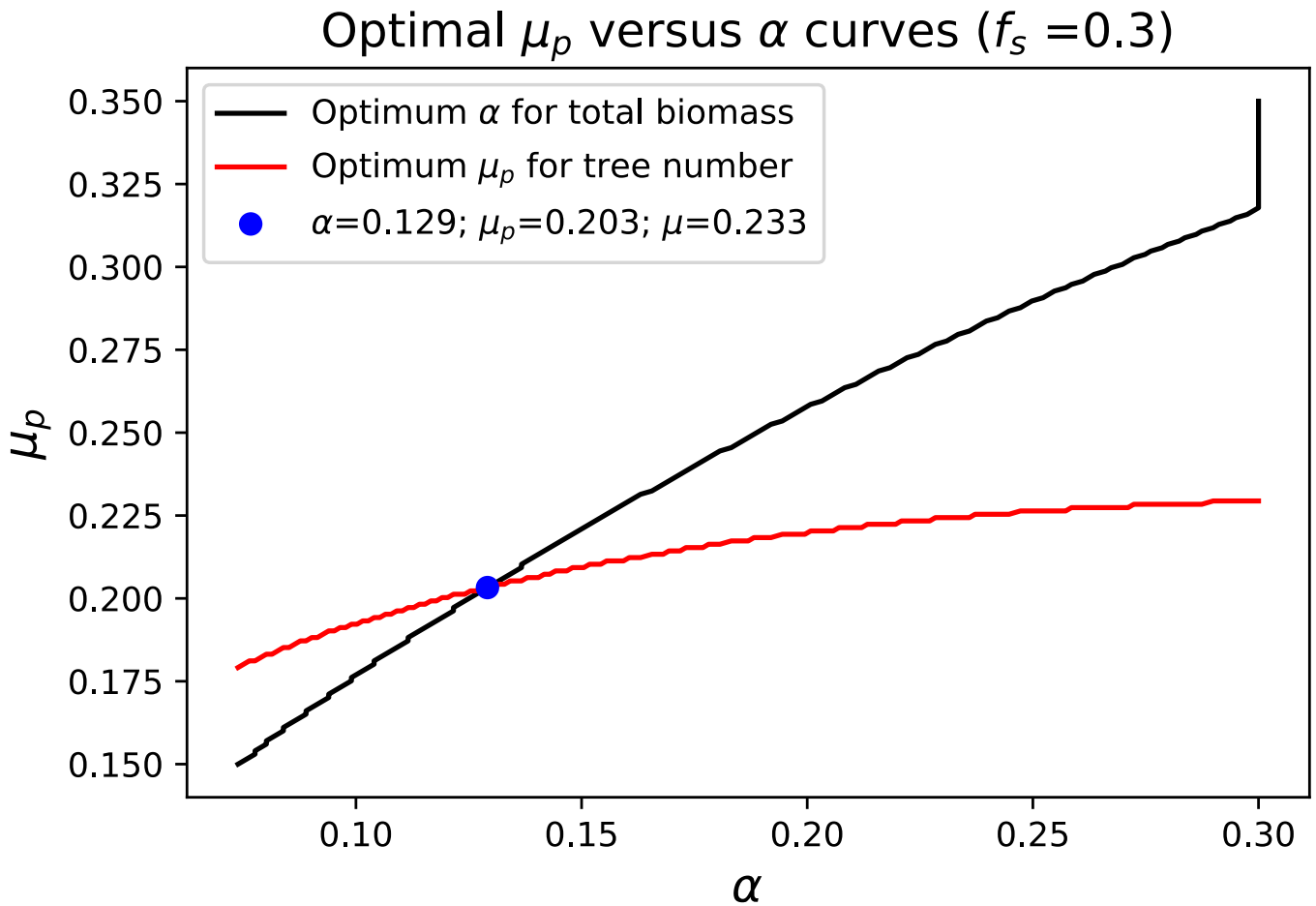
Our ***Demographic Optimality*** Hypothesis

Forests simultaneously maximize:

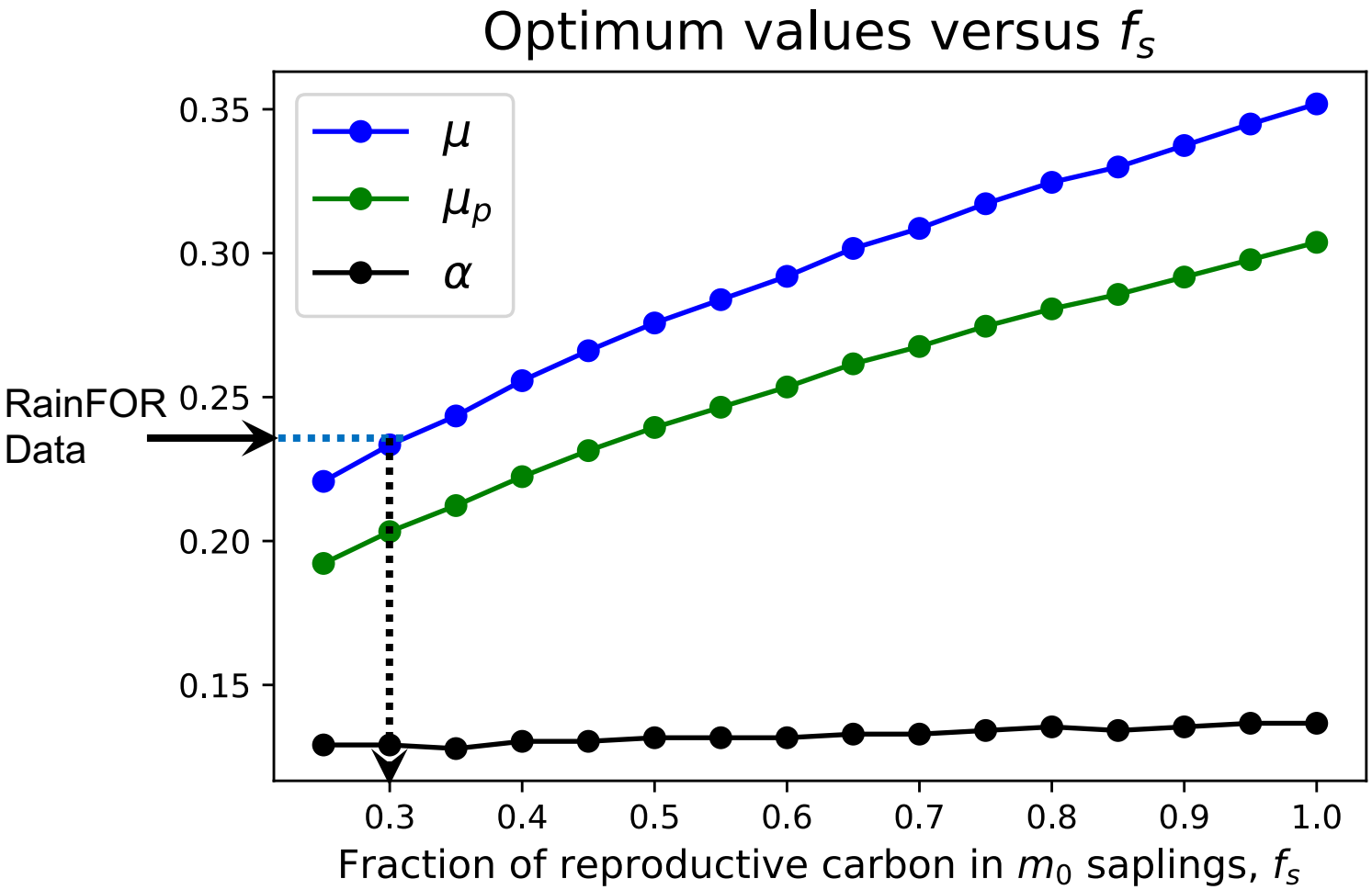
a) Tree number as a function of the ratio of mortality to productivity (μ_p), for a given allocation of assimilate to reproduction (α);

b) Total biomass as a function of the allocation of assimilate to reproduction (α), for a given ratio of mortality to productivity (μ_p).

Demographic Optimality for f_s=0.3



Dependence of Demographic Optimality on f_s



Conclusions

- Metabolic scaling theory (MST) of tree growth-rates, plus the assumption of demographic equilibrium (DET), reproduces tree-size distributions remarkably well;
- Observed tree-size in forests worldwide suggest a near common ratio of mortality to growth rate for a 1kg sapling (μ_1 = 0.2 to 0.25);
- This is consistent with simultaneous maximization of tree number and total biodiversity (which we call ***'demographic optimality***.